

Gamma controls readout learning speed

A small-gamma network can represent a target and still take far longer to learn it. Gamma smooths the features, suppressing fine spatial variation. This changes the readout kernel. Once we calculate that kernel, classical gradient-descent theory predicts the learning curve and its crossing of an accuracy threshold.

Freeze the centers c_j and common slope $\gamma > 0$, and train only the readout from zero:

$$f_{\gamma, \theta}(x) = b + \sum_{j=1}^w w_j \tanh(\gamma(x - c_j)).$$

Where gamma enters. Each tanh is a sharp step smoothed over a distance proportional to $1/\gamma$. We show on page 2 that this smoothing multiplies frequency ω by

$$M_{\gamma}(\omega) = \frac{z}{\sinh z}, \quad z = \frac{\pi|\omega|}{2\gamma}, \quad M_{\gamma}(0) = 1. \quad (1)$$

For $|\omega| \gg \gamma$, the multiplier is exponentially small. Applying the smoothing to both factors of a feature inner product gives the gamma-dependent kernel exactly.

Where the learning-time prediction comes from. Let Φ_{γ} be these features, including the bias, evaluated at m training inputs, and set $K_{\gamma} = \Phi_{\gamma} \Phi_{\gamma}^T / m$. For target vector $y \neq 0$, half mean squared error, and step $0 < \eta_{\gamma} \leq 1/\|K_{\gamma}\|_2$, classical GD theory [1] gives the relative residual

$$E_{\gamma}(n) = \frac{\|(I - \eta_{\gamma} K_{\gamma})^n y\|_2}{\|y\|_2}. \quad (2)$$

The predicted time to 1% error is simply the first n for which this curve falls below 0.01. This calculation uses the kernel and target, without fitting a training trajectory. The finite center and sample geometry is retained; Fourier frequencies need not be kernel eigenvectors.

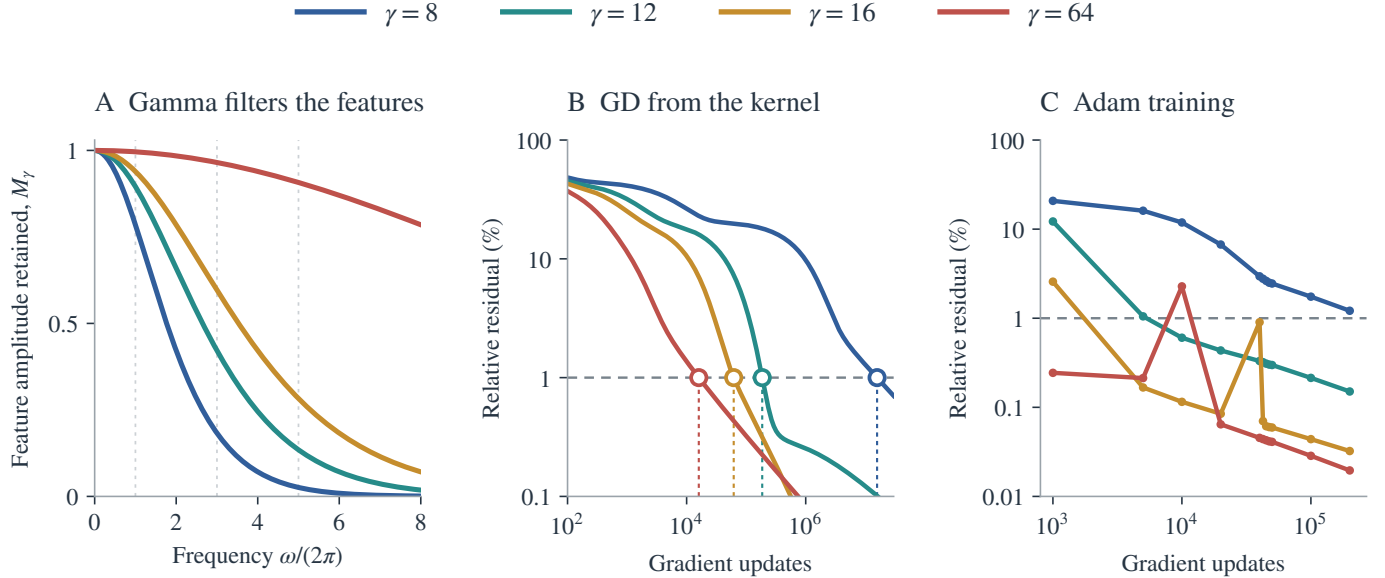


Figure 1: Same target, different access during optimization. The target is $\sin(2\pi x) + \frac{1}{2} \sin(6\pi x) + \frac{1}{4} \sin(10\pi x)$, with 559 fixed-center features and 8193 training inputs on $[-1, 1]$. **A:** The exact gamma filter; vertical guides mark the target frequencies. **B:** Solid curves are kernel predictions; open circles mark executed GD first hits of 1%. Dotted verticals project predicted crossings onto update time. **C:** Adam learning curves join saved checkpoints, with identical optimizer settings across gammas. Dashed horizontals mark 1%. Both optimizers start at zero; GD uses $\eta_{\gamma} \simeq 0.5/\|K_{\gamma}\|_2$.

For this target, gamma 8 takes **15,798,313 GD updates** to reach 1%, versus **16,013** at gamma 64. The predicted delay ratio is **985.70–987.49**; the measured ratio is **986.59**. The approximation bounds certify the threshold crossings shown in B. Both models reach 1%, so this is an optimization delay at attainable accuracy. Adam also shows a substantial separation with the same features (C); its curves are empirical, while (2) applies to GD.

Proof of the gamma-to-kernel statement

Let $\rho_\gamma(t) = \gamma \operatorname{sech}^2(\gamma t)/2$. This density has unit mass. Splitting at a step's center gives

$$\int_{\mathbb{R}} \rho_\gamma(x-t) \operatorname{sign}(t-c) dt = 2 \int_{-\infty}^{x-c} \rho_\gamma(u) du - 1 = \tanh(\gamma(x-c)).$$

Define the step kernel using the same centers,

$$k_{\text{step}}(t, s) = 1 + \sum_j \operatorname{sign}(t - c_j) \operatorname{sign}(s - c_j).$$

Taking feature inner products gives

$$\begin{aligned} k_\gamma(x, x') &= 1 + \sum_j \tanh(\gamma(x - c_j)) \tanh(\gamma(x' - c_j)) \\ &= \iint \rho_\gamma(x-t) \rho_\gamma(x'-s) k_{\text{step}}(t, s) dt ds. \end{aligned}$$

The bias is preserved by unit mass. Bounded features justify exchanging the finite sum and integrals. Sampling this identity yields $(K_\gamma)_{i\ell} = k_\gamma(x_i, x_\ell)/m$.

For the Fourier convention $\widehat{\rho}(\omega) = \int \rho(t) e^{-i\omega t} dt$, set $a = \omega/(2\gamma)$ and $v = e^{2\gamma t}$. The beta integral gives

$$\widehat{\rho}_\gamma(\omega) = \int_0^\infty \frac{v^{-ia}}{(1+v)^2} dv = \Gamma(1-ia)\Gamma(1+ia) = \frac{\pi a}{\sinh(\pi a)} = M_\gamma(\omega).$$

Here Γ is Euler's gamma function. The final equality follows from $\Gamma(1+ia) = ia\Gamma(ia)$ and the reflection identity $\Gamma(ia)\Gamma(1-ia) = \pi/\sin(\pi ia)$; the value at zero is one by continuity. Convolution therefore multiplies the wave $e^{i\omega x}$ by $M_\gamma(\omega)$ in either kernel argument.

For $z > 0$, the derivative of $z/\sinh z$ has numerator $\sinh z - z \cosh z < 0$: it starts at zero and its derivative is $-z \sinh z < 0$. Consequently, a cap $\gamma \leq \bar{\gamma}$ bounds the multiplier by $M_{\bar{\gamma}}(\omega)$. Also, $\sinh z \sim e^z/2$ gives $M_\gamma(\omega) \sim 2ze^{-z}$. This proves the explicit attenuation and its dependence on gamma, with no periodic assumption. $\square$

Writing the residual as $r_n = y - \Phi_\gamma \theta_n$, the standard GD identity follows from $r_{n+1} = (I - \eta_\gamma K_\gamma) r_n$ and $r_0 = y$. Diagonalization gives a factor $(1 - \eta_\gamma \lambda)^n$ in each eigenmode of eigenvalue λ . Thus gamma changes the learning rates through the kernel, and the target's energy in those directions determines the delay. The cap bounds attenuation; the numerical delay is specific to the target and geometry.

[1] Y. Yao, L. Rosasco, A. Caponnetto. [On Early Stopping in Gradient Descent Learning](#). *Constructive Approximation* 26:289–315, 2007, §3.3.

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